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### Electric oil pump

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#### Abstract

An electric oil pump comprises a stator assembly and an electronic control panel assembly which are electrically connected. The electronic control panel assembly comprises an electronic component. The electric oil pump further comprises a conductive pump housing and a conductive member that electrically connects a reference ground layer of the electronic control panel assembly and the pump housing. The above arrangement facilitates reducing electromagnetic radiation of the electronic control panel assembly, thereby reducing interference of the electromagnetic radiation of the electronic control panel assembly on the electronic component and/or other external devices.

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| <b>Family ID:</b>            | <b>80539267</b>                                                                                                   |
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## Background/Summary

(1) The present application is a National Phase entry of PCT Application No. PCT/CN2021/118180, filed on Sep. 14, 2021, which claims the benefit of priorities to the following two Chinese patent applications, which are incorporated herein by reference in their entireties, 1) Chinese Patent Application No. 202010970515.X, titled “ELECTRIC OIL PUMP”, filed with the China National Intellectual Property Administration on Sep. 15, 2020 and 2) Chinese Patent Application No. 202011057359.4, titled “ELECTRIC OIL PUMP”, filed to the China National Intellectual Property Administration on Sep. 29, 2020, the application date is Sep. 15, 2020.

### FIELD

(2) The present application relates to the technical field of vehicles, and in particular to an assembly of a lubrication system and/or a cooling system of a vehicle.

### BACKGROUND

(3) An electronic oil pump mainly provides power source for a lubrication system and/or cooling system of a vehicle; the electronic oil pump includes an electric control board assembly, the electric control board assembly includes an electronic component; during the use of the electronic oil pump, the electric control board assembly may generate electromagnetic radiation, these electromagnetic radiation may cause interference to electronic components and/or other external devices, thus affecting the performance or use of electronic components and/or other external devices.

### SUMMARY

(4) An object according to the present application is to provide an electronic oil pump, which may reduce an electromagnetic radiation of an electric control board assembly, and further reduce the interference of the electromagnetic radiation of the electric control board assembly to electronic components and/or other external devices.

(5) In order to achieve the above object, the following technical solution is provided according to an embodiment of the present application.

(6) An electronic oil pump includes a stator assembly and an electric control board assembly, the stator assembly is electrically connected with the electric control board assembly, the electric control board assembly includes an electronic component; the electronic oil pump further includes a pump housing and a conductive member, the pump housing can conduct electricity, and the conductive member electrically connects a reference ground plane of the electric control board

assembly with the pump housing.

(7) In this way, because the conductive member is electrically connected with the reference ground plane of the electric control board assembly and the pump housing, a low impedance path can be provided for the interference voltage in the electric control board assembly, which is beneficial to reducing an electromagnetic radiation of an electric control board assembly, and further to reducing the interference of the electromagnetic radiation of the electric control board assembly to electronic components and/or other external devices.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional structural view of a first embodiment of an electronic oil pump provided according to the present application;
- (2) FIG. 2 is a schematic front view of the electronic oil pump in FIG. 1 with a pump cover removed;
- (3) FIG. 3 is a three-dimensional structure diagram of a partition member and a conductive member in FIG. 1 assembled together;
- (4) FIG. 4 is a schematic diagram of a front structure of the partition member and the conductive member assembled together in FIG. 3;
- (5) FIG. 5 is a schematic structural diagram of a section taken along the A-A direction in FIG. 4;
- (6) FIG. 6 is a three-dimensional structure diagram of an electric control board assembly and the conductive member in FIG. 1;
- (7) FIG. 7 is a schematic diagram of a front structure of the electric control board assembly and the conductive member in FIG. 6;
- (8) FIG. 8 is a schematic structural diagram of a section taken along the A-A direction in FIG. 6;
- (9) FIG. 9 is a schematic three-dimensional structure diagram of the conductive member in FIG. 1;
- (10) FIG. 10a is a three-dimensional structure diagram of a first conductive part in FIG. 9;
- (11) FIG. 10b is an enlarged structural diagram of part A of the first conductive part in the left figure;
- (12) FIG. 11 is a three-dimensional structure diagram of a second conductive part in FIG. 9;
- (13) FIG. 12 is a schematic cross-sectional structural view of a second embodiment of the electronic oil pump provided according to the present application;
- (14) FIG. 13 is a three-dimensional structure diagram of the partition member, the conductive member and a shielding member in FIG. 12 assembled together;
- (15) FIG. 14 is a front view of the partition member, the conductive member and the shielding member in FIG. 13 assembled together;
- (16) FIG. 15 is a schematic structural diagram of a section taken along the A-A direction in FIG. 14;
- (17) FIG. 16 is a schematic three-dimensional structure diagram of the shielding member in FIG. 12;
- (18) FIG. 17 is a schematic three-dimensional structure diagram of the conductive member in FIG. 12;
- (19) FIG. 18 is a schematic cross-sectional structural view of a third embodiment of the electronic oil pump provided according to the present application;
- (20) FIG. 19 is a three-dimensional structure diagram of the partition member, the conductive member and a shielding member in FIG. 18 assembled together;
- (21) FIG. 20 is a schematic cross-sectional structural view of a fourth embodiment of the electronic oil pump provided according to the present application;
- (22) FIG. 21 is a three-dimensional structure diagram of the partition member, the conductive

member and a shielding member in FIG. 20 assembled together; and

(23) FIG. 22 is a schematic cross-sectional structural view of a fifth embodiment of the electronic oil pump provided according to the present application.

#### DETAILED DESCRIPTION OF THE EMBODIMENTS

(24) The present application is further described as follows in conjunction with the drawings and specific embodiments.

(25) The electronic oil pump in the following embodiment mainly provide flowing power for the working medium of the lubrication system and/or cooling system of a vehicle, and specifically provide flowing power for the working medium of the lubrication system and/or cooling system in a transmission system of the vehicle.

(26) Referring to FIG. 1, an electronic oil pump **100** includes a pump housing, a first rotor assembly **2**, a stator assembly **4**, a second rotor assembly **3** and an electric control board assembly **6**; the pump housing forms a pump cavity, and, the first rotor assembly **2**, the stator assembly **4**, the second rotor assembly **3** and the electric control board assembly **6** are arranged in the pump cavity, in this embodiment, the pump cavity includes a first cavity **70**, a second cavity **80** and a third cavity **90**, and, the first cavity **70**, the second cavity **80** and the third cavity **90** are sequentially distributed along the height direction, the first rotor assembly **2** is arranged in the first cavity **70**, the stator assembly **4** and the second rotor assembly **3** are arranged in the second cavity **80**, the electric control board assembly **6** is arranged in the third cavity **90**. The first cavity **70** is communicated with the second cavity **80**, and the second cavity **80** is not communicated with the third cavity **90**; the stator assembly **4** includes a stator iron core **41**, an insulating frame **42** and a winding **43**, and, the insulating frame **42** covers at least part of a surface of the stator iron core **41**, and the winding **43** is wound around the insulating frame **42**; when the electronic oil pump **100** operates, the electric control board assembly **6** controls the current in the winding **43** passing through the stator assembly **4** to change according to a predetermined rule, thereby controlling the stator assembly **4** to generate a changing excitation magnetic field, the second rotor assembly **3** rotates under the action of the excitation magnetic field, and the second rotor assembly **3** can directly or indirectly drive the first rotor assembly **2** to rotate; when the first rotor assembly **2** rotates, a volume of the hydraulic chamber between the first rotor and the second rotor of the first rotor assembly **2** changes, so that the working medium is pressed out to the outlet to generate flowing power; in this embodiment, at least part of the working medium in the first cavity **70** can flow into the second cavity **80**, the working medium in the second cavity **80** can cool the stator assembly **4** since the stator assembly **4** is arranged in the second cavity **80**, facilitating the heat dissipation of the stator assembly **4**.

(27) Referring to FIG. 1, in this embodiment, the pump housing includes a pump cover **1**, a first housing **7** and a second housing **8**, and, the pump cover **1** is fixedly connected with the first housing **7** and the first housing **7** is fixedly connected with the second housing **8**; specifically, in this embodiment, the pump cover **1** is connected with the first housing **7** by screws or bolts, this arrangement makes the disassembly and assembly of the electronic oil pump more convenient, thus facilitating the maintenance of the first rotor assembly **2** of the electronic oil pump, of course, the pump cover **1** and the first housing **7** may also be connected by other ways, such as plugging and clamping; the first housing **7** is fixedly connected with the second housing **8**, specifically, the first housing **7** and the second housing **8** are connected by screws or bolts, so that the electronic oil pump is more convenient to assemble and disassemble, in this embodiment, the electric control board assembly **6** is arranged in the cavity between the first housing **7** and the second housing **8**, which is beneficial to the maintenance of the electric control board assembly in the electronic oil pump, and the connection between the first housing **7** and the second housing **8** is relatively reliable, of course, the first housing **7** and the second housing **8** may also be connected by plugging, clamping or other means.

(28) Referring to FIG. 1 and FIG. 2, the first rotor assembly **2** includes a first rotor **21** and a second rotor **22**, the first rotor **21** includes multiple inner teeth, and the second rotor **22** includes multiple

outer teeth, a hydraulic cavity **801** is formed between the inner teeth of the first rotor **21** and the outer teeth of the second rotor **22**, in this embodiment, the hydraulic cavity is also a part of the first cavity **70**, in this embodiment, the first rotor **21** is sleeved on an outer periphery of the second rotor **22**. Referring to FIG. **1** again, the electronic oil pump further includes an inlet **11** and an outlet (not shown), the working medium can enter the hydraulic cavity **801** through the inlet **11** and leave the hydraulic cavity **801** through the outlet (not shown); as there is a certain eccentricity between the first rotor **21** and the second rotor **22**, some outer teeth of the second rotor **22** mesh with some inner teeth of the first rotor **21** when the second rotor **22** rotates, thus driving the first rotor **21** to rotate; during one revolution of the first rotor **21** and the second rotor **22**, an internal volume of the hydraulic cavity **801** changes, specifically, when the first rotor assembly **2** rotates to a certain angle from the initial position, the volume in the hydraulic cavity **801** gradually increases to form a partial vacuum, the working medium is sucked into the hydraulic cavity **801** from the inlet **11**, when the first rotor **21** and the second rotor **22** continue to rotate, the volume of the hydraulic cavity **801** filled with the working medium gradually decreases, and the working medium is squeezed, so that the working medium entering the hydraulic cavity **801** is pressed out to the outlet (not shown), thereby generating flowing power; in this embodiment, the electronic oil pump **100** further includes a pump shaft **15**, which can drive part of the first rotor assembly **2** to rotate, specifically, in this embodiment, the pump shaft **15** can drive the second rotor **22** to rotate, in this embodiment, the pump shaft **15** is connected with the second rotor **22**, and the pump shaft **15** is connected with the second rotor assembly **3**, the second rotor assembly **3** drives the second rotor **22** to rotate through the pump shaft **15**, thereby realizing the rotation of the first rotor assembly **2**.

(29) Referring to FIG. **1**, FIG. **1** is a structural diagram of a first embodiment of an electronic oil pump in the present application; the structure of the electronic oil pump of the first embodiment is described in detail below.

(30) Referring to FIG. **1**, the electronic oil pump **100** further includes a conductive member **9**, the conductive member **9** electrically connects the reference ground plane of the electric control board assembly **6** with the first housing **7**, and the first housing **7** can conduct electricity; at least part of the conductive member **9** is located in the inner cavity of the first housing **7**, the material of the first housing **7** is metal, and the first housing **7** partially surrounds the periphery of the stator assembly **4**; the stator assembly **4** is located in the inner cavity of the first housing **7**, and when the winding **43** of the stator assembly **4** radiates electromagnetic waves outward or when an external system radiates electromagnetic waves to the first housing **7**, on one hand, the electromagnetic wave radiated by the winding **43** can be absorbed and reflected by the first housing **7** itself, which is beneficial to preventing the electromagnetic wave radiated by the winding **43** from affecting the external system, on the other hand, the electromagnetic wave radiated by the external system can be absorbed and reflected by the first housing **7** itself, which is beneficial to preventing the electromagnetic wave radiated by the external system from affecting the performance of the electronic oil pump; specifically, referring to FIG. **1**, the conductive member **9** includes a first part **911** and a second part **921**, the first part **911** is electrically connected with the reference ground plane of the electric control board assembly **6**, the second part **921** abuts against the inner wall of the first housing **7**, the reference ground plane of the electric control board assembly **6** is electrically conducted with the first housing **7** through the conductive member **9**; in this way, since the first housing **7** is electrically connected with the reference ground plane of the electric control board assembly **6**, the beneficial effects are: firstly, a low-impedance path is provided for the interference voltage in the electric control board assembly **6**, which is conducive to reducing the electromagnetic radiation of the electric control board assembly **6**, and further beneficial to reducing the interference of the electromagnetic radiation of the electric control board assembly **6** to electronic components and/or other external devices; secondly, when the surface of the first housing **7** has static electricity, the static electricity can be transmitted to the reference ground plane of the electric control board assembly **6** through the conductive member **9**, when the reference

ground plane of the electric control board assembly 6 is grounded to the outside, the static electricity on the reference ground plane is conducted to the outside grounding point, which is conducive to reducing the static electricity accumulated on the surface of the first housing 7, and thus to reducing the impact of the above static electricity on the performance of the external system and/or the electric control board assembly 6, thirdly, when the surface of the electric control board assembly 6 has static electricity, the above static electricity can be conducted to the first housing 7 through the conductive member 9, which is conducive to reducing the static electricity accumulated on the surface of the electric control board assembly 6, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components; besides, in this embodiment, the pump cover 1, the first housing 7 and the second housing 8 are made of metal, and, the stator core 41, the second housing 8 and the pump cover 1 are all in contact with the first housing 7, so that the stator core 41 and the second housing 8 are electrically connected with the reference ground plane of the electric control board assembly 6, on one hand, the static electricity on the surface of the stator core 41, the second housing 8 and the pump cover 1 can be conducted to the reference ground plane of the electric control board assembly 6, the above static electricity is conducted to the outside grounding point, which is conducive to reducing the influence of static electricity on the electric control board assembly 6 and/or other external devices; on the other hand, the grounding area of the electronic oil pump can be increased, which can provide a low impedance path for the interference voltage in the electric control board assembly 6 and the stator assembly 4, which is beneficial to reducing an electromagnetic radiation of the electric control board assembly 6 and the stator assembly 4, and further to reducing the interference of the above electromagnetic radiation to electronic components and/or other external devices; in this embodiment, the pump cover 1, the first housing 7 and the second housing 8 are made of metal, of course, it is also possible that the material of the first housing 7 is metal, and the material of the pump cover 1 and the second housing 8 is nonmetal. Referring to FIG. 1, in this embodiment, one end of the conductive member 9 contacts with the first housing 7, but of course, one end of the conductive member 9 may also contact the stator core 41 or the second housing 8.

(31) Referring to FIG. 1 and FIG. 3, the electronic oil pump 100 further includes a partition member 5, at least part of the partition member 5 is arranged between the stator assembly 4 and the electric control board assembly 6, the second cavity 80 is located at one side of the main body 51 of the partition member 5, the third cavity 90 is located on the other side of the main body 51 of the partition member 5; in this embodiment, part of the conductive member 9 is fixedly connected with the main body 51 of the partition member 5, the conductive member 9 passes through the main body 51 of the partition member 5 in the vertical direction of the electronic oil pump 100, specifically, in this embodiment, the conductive member 9 is used as an insert, and the partition member 5 is formed by injection molding, that is, the conductive member 9 and the partition member 5 are fixedly connected by injection molding, the joint between the conductive member 9 passing through the partition member 5 and the partition member 5 is sealed; specifically, referring to FIG. 3 to FIG. 5, the partition member 5 includes a first convex part 52, the first convex part 52 protrudes from the upper surface of the main body 51 of the partition member 5, and part of the conductive member 9 passes through the first convex part 52; the first convex part 52 includes a groove 521, and the groove 521 is recessed from the upper surface of the first convex part 52, the groove 521 does not reach the lower surface of the main body 51 of the partition member 5, the conductive member 9 passing through the main body 51 of the partition member 5 passes through the groove 521, there is a gap between the outer periphery of the conductive member 9 located in the groove 521 and the inner wall of the groove 521, and the gap is filled with sealant, the sealant helps to prevent the working medium in the second cavity 80 from leaking to the third cavity 90 through the joint between the conductive member 9 and the partition member 5, so as not to affect the performance of the electric control board assembly 6 located in the third cavity 90.

(32) Referring to FIG. 6 to FIG. 8, the electric control board assembly 6 includes a substrate 61, the

substrate **61** includes a first surface **615** and a second surface **616**, the first surface **615** is closer to the stator assembly **4** than the second surface **616**; the substrate **61** includes a through hole **611**, the through hole **611** is defined in the thickness direction of the substrate **61**, the first part **911** of the conductive member **9** inserts into the through hole **611** and is tightly matched with the through hole **611**; in this embodiment, the first part **911** of the conductive member **9** is connected with the substrate **61** by providing the through hole **611** in the substrate **61**, of course, it is also possible to connect the first part **911** of the conductive member **9** with the substrate **61** by providing an abutting part **613** on the electric control board assembly **6**, the abutting part **613** is located on the first surface **615**, the abutting part **613** is electrically connected with the reference ground plane of the electric control board assembly **6**, the first part **911** is in contact with the abutting part **613** and electrically connected with the abutting part **613**, the “abutting part” here may be a separate conductive part, which is fixedly connected with the substrate **61**, or it may be a conductive layer, the conductive layer can be treated by tin plating, chemical nickel plating and gold immersion.

(33) Referring to FIG. **9** to FIG. **10**, in this embodiment, the conductive member **9** is at least partially elastic, and the elastic conductive member **9** includes an inclined part **922**, the front end of the inclined part **922** is connected with the second part **921**, or, the second part **921** is located at the front end of the inclined part **922**, the inclined part **922** is integrally provided with the second part **921**; in combination with FIG. **1**, the inclined part **922** is disposed at an angle with the inner wall of the first housing **7**, the first housing **7** exerts a force on the inclined part **922**, which is beneficial to improving the reliability of the contact between the second part **921** and the first housing **7**.

(34) Specifically, referring to FIG. **9** to FIG. **11**, the conductive member **9** includes a first conductive part **91** and a second conductive part **92**, and in this embodiment, the first conductive part **91** and the second conductive part **92** are separately arranged, the “separate arrangement” here means that the first conductive part **91** and the second conductive part **92** are processed separately and then assembled; of course, the first conductive part **91** and the second conductive part **92** may also be an integral structure, the “integral structure” here means that the first conductive part **91** and the second conductive part **92** are processed into one piece; referring to FIG. **9** to FIG. **11**, in this embodiment, the first part **911** is formed on the first conductive part **91**, the second part **921** and the inclined part **922** are formed on the second conductive part **92**, the second conductive part **92** has elasticity, and the first conductive part **91** is in contact with and connected with the second conductive part **92**; in combination with FIG. **1**, FIG. **9** to FIG. **11**, in this embodiment, one end of the first conductive part **91** is electrically connected with the reference ground plane of the electric control board assembly **6**, the other end of the first conductive part **91** is connected with the insulating frame **42**, the first conductive part **91** is non-electrically connected with the winding **43** in FIG. **1**.

(35) Referring to FIG. **9** to FIG. **11**, in this embodiment, the second conductive part **92** further includes a body part **923** and a limiting part **924**, part of the first conductive part **91** is in contact with the body part **923**, one end of the limiting part **924** is connected with the first side surface **9231** of the body part **923**, the other end of the limiting part **924** is connected with the second side surface **9232** of the body part **923**, the inclined part **922** is connected with the upper end of the body part **923**, in this embodiment, the inclined part **922** is connected with the upper end of the body part **923** by an arc part, the second part **921** is located at one side of the body part **923**, and part of the limiting part **924** is located at the other side of the body part **923**; the limiting part **924** has an accommodating cavity **9240**, the first conductive part **91** passes through the accommodating cavity **9240**, in combination with FIG. **9**, the first conductive part **91** passes through the limiting part **924**, one end of the first conductive part **91** is located at one side of the limiting part **924**, one end of the second conductive part **92** is located at the other side of the limiting part **924**; referring to FIG. **9** to FIG. **11**, the first conductive part **91** includes a first limiting surface **912**, the lower end surface **9241** of the limiting part **924** of the second conductive part **92** is located above the first limiting surface **912**, or the lower end surface **9241** of the limiting part **924** abuts against the first

limiting surface **912**, and the second limiting part **924** can be prevented from sliding down along the length direction of the first conductive part **91**.

(36) Referring to FIG. **10**, in this embodiment, the first conductive part **91** further includes a mating surface **913** and a second limiting surface **914**, one end of the first limiting surface **912** is connected with one end of the mating surface **913**, one end of the second limiting surface **914** is connected with the other end of the mating surface **913**, in combination with FIG. **9** to FIG. **11**, a notch is formed on the side surface of the first conductive part **91**, the first limiting surface **912**, the mating surface **913** and the second limiting surface **914** are respectively formed on the wall of the notch from bottom to top, at least part of the mating surface **913** is located in the accommodating cavity **9240** of the limiting part **924**, the first limiting surface **912** is located at one side of the limiting part **924**, and the second limiting surface **914** is located at the other side of the limiting part **924**, and the second limiting part **924** can be prevented from sliding up and down along the length direction of the first conductive part **91**; referring to FIG. **10**, the first conductive part **91** further includes an inclined surface **915**, one end of the inclined surface **915** is connected with the other end of the second limiting surface **914**, along the length direction of the first conductive part **91**, the inclined surface **915** is located above the mating surface **913**, referring to FIG. **11**, the limiting part **924** further includes a first sub-part **9242** and a second sub-part **9243**, the first end of the first sub-part **9242** is connected with the first side surface **9231** of the body part **923**, the second end **9244** of the first sub-part **9242** is a free end, the first end of the second sub-part **9243** is connected with the second side surface **9232** of the body part **923**, the second end **9245** of the second sub-part **9243** is a free end, the second end **9244** of the first sub-part **9242** has a set distance from the second end **9245** of the second sub-part **9243**, referring to FIG. **9** to FIG. **11** in combination, when the limiting part **924** of the second conductive part **92** is inserted from the upper end of the first conductive part **91**, the side wall corresponding to the accommodating cavity **9240** of the limiting part **924** is in contact with the inclined surface **915** of the first conductive part **91**, and the first sub-part **9242** and the second sub-part **9243** can be elastically deformed, so that the mating surface **913** can be located in the accommodating cavity **9240** of the limiting part **924**, thereby realizing the limiting connection between the second conductive part **92** and the first conductive part **91**; in this embodiment, after the second conductive part **92** is limitedly connected with the first conductive part **91**, the second conductive part **92** and the first conductive part **91** are fixed by welding.

(37) Referring to FIG. **12**, FIG. **12** is a structural diagram of a second embodiment of an electronic oil pump in the present application; the structure of the electronic oil pump of the second embodiment is described in detail below.

(38) Referring to FIG. **12**, in this embodiment, the electronic oil pump **100a** further includes a shielding member **10**, the stator assembly **4**, the shielding member **10** and the electric control board assembly **6** are axially distributed along the height direction of the electronic oil pump, at least part of the shielding member **10** is located between the stator assembly **4** and the electric control board assembly **6**; the shielding member **10** is made of conductive metal material, and at least part of the shielding member **10** covers at least part of the winding **43**; in this way, the shielding member **10** can absorb and reflect at least part of the electromagnetic waves radiated by the winding **43** toward the electric control board assembly **6**, which is beneficial to reducing the interference of the electromagnetic waves generated by the winding **43** on the electric control board assembly **6**.

(39) Referring to FIG. **12** and FIG. **13**, in this embodiment, the shielding member **10** is fixedly connected with the partition member **5**, of course, the shielding member **10** and the partition member **5** can also be limited; specifically, in this embodiment, the shielding member **10** is located between the stator assembly **4** and the main body **51** of the partition member **5**, the shielding member **10** is fixedly connected with the main body **51** of the partition member **5**, of course, the shielding member **10** may also be limited with the main body **51** of the partition member **5**.

(40) Specifically, referring to FIG. **13** to FIG. **16**, the partition member **5** further includes at least two protruding posts **53**, along the axial direction of the partition member **5**, the at least two

protruding posts **53** is convexly provided from the upper end face of the main body **51** of the partition member **5**; the shielding member **10** includes at least two through holes **101**, and the at least two protruding posts **53** passes through the through holes **101**, the shielding member **10** and the partition member **5** can be fixedly connected by hot riveting the part of the at least two protruding posts **53** protruding out of the through hole **101**; of course, the partition member **5** can also be formed by injection molding with the shielding member **10** as an insert, that is, the shielding member **10** is connected to the partition member **5** by injection molding.

(41) Referring to FIG. **12** to FIG. **17**, the electronic oil pump **100a** further includes a conductive member **9**, the conductive member **9** passes through the main body **51** of the partition member **5**, the conductive member **9** includes a first part **911** and an upper part **925**, the first part **911** is electrically connected with the reference ground plane of the electric control board assembly **6**, in this embodiment, the upper part **925** of the conductive member **9** is in contact with the shielding member **10**, the conductive member **9** is electrically connected with the reference ground plane of the electric control board assembly **6** and the shielding member **10**; in the above way, the shielding member **10** is indirectly electrically connected with the reference ground plane of the electric control board assembly **6**; the beneficial effects are: firstly, a low impedance path is provided for the interference voltage in the stator assembly **4**, which is beneficial to reducing the electromagnetic radiation of the stator assembly **4**, and further reducing the interference of electromagnetic radiation generated by stator assembly **4** on electric control board assembly **6** and/or other external devices; secondly, when the surface of the shielding member **10** has static electricity, the static electricity can be conducted to the reference ground plane of the electric control board assembly **6**, when the reference ground plane of the electric control board assembly **6** is grounded to the outside, for example, the reference ground plane of the electric control board assembly **6** is grounded to the outside through the electric connection between the conductive member and the pump housing; or the reference ground plane of the electric control board assembly **6** is grounded to the outside by other means, the static electricity conducted to the reference ground plane of the electric control board assembly **6** is then conducted to the outside grounding point, which is conducive to reducing the static electricity accumulated on the surface of the shielding member **10**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components; thirdly, when the surface of the electric control board assembly **6** has static electricity, the above static electricity can be conducted to the shielding member **10** through the conductive member **9**, which is conducive to reducing the static electricity accumulated on the surface of the electric control board assembly **6**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components.

(42) Referring to FIG. **12**, in this embodiment, the upper part **925** of the conductive member **9** is connected with the insulating frame **42**, and the conductive member **9** is non-electrically connected with the winding **43**, of course, the top end of the upper part **925** of the conductive member **9** may also be a free end, in which case the upper part **925** of the conductive member **9** is not connected with the insulating frame **42** or other components.

(43) Referring to FIG. **16**, the shielding member **10** includes a base part **102** and a lead-out part **103**, the lead-out part **103** is connected to the peripheral surface of the base part **102**, and the lead-out part **103** is convex; specifically, in this embodiment, the base part **102** and the lead-out part **103** are of an integral structure, of course, the base part **102** and the lead-out part **103** may be solely arranged and fixedly connected; referring to FIG. **15** and FIG. **16**, in this embodiment, the lead-out part **103** is in contact with the upper part **925** of the conductive member **9**, and the lead-out part **103** is elastic, the lead-out part **103** includes an inclined section **1031**, the main section of the inclined section **1031** is arranged at an angle with the conductive member **9**, and the front end of the inclined section **1031** is in contact with the conductive member **9**, which helps to improve the reliability of the contact between the lead-out part **103** and the conductive member **9**.

(44) Compared with the first embodiment of the electronic oil pump, in this embodiment, the electronic oil pump **100a** further includes a shielding member **10**, the conductive member **9** is electrically connected with the reference ground plane of the electric control board assembly **6** and the shielding member **10**; other features in this embodiment can refer to the first embodiment of the electronic oil pump, which is not be described here.

(45) Referring to FIG. **18**, FIG. **18** is a structural diagram of a third embodiment of an electronic oil pump in the present application; the structure of the electronic oil pump of the third embodiment is described in detail below.

(46) Referring to FIG. **18** and FIG. **19**, in this embodiment, the electronic oil pump **100b** further includes a shielding member **10**, the shielding member **10** is located between the main body **51** of the partition member **5** and the electric control board assembly **6**, the shielding member **10** is fixedly connected with the main body **51** of the partition member **5**, of course, the shielding member **10** may also be limited with the main body **51** of the partition member **5**; specifically, referring to FIG. **19**, the partition member **5** further includes at least two protruding posts **53**, along the axial direction of the partition member **5**, the at least two protruding posts **53** are convexly provided from the lower end face of the main body **51** of the partition member **5**; the shielding member **10** includes at least two through holes, and the at least two protruding posts **53** passes through the through holes, the shielding member **10** and the partition member **5** can be fixedly connected by hot riveting the part of the at least two protruding posts **53** protruding out of the through hole; of course, the partition member can also be formed by injection molding with the shielding member **10** as an insert, that is, the shielding member **10** is connected to the partition member **5** by injection molding.

(47) Compared with the second embodiment of the electronic oil pump, in this embodiment, the shielding member **10** is located between the main body **51** of the partition member **5** and the electric control board assembly **6**; other features in this embodiment can refer to the second embodiment of the electronic oil pump, which is not be described here.

(48) Referring to FIG. **20**, FIG. **20** is a structural diagram of a fourth embodiment of an electronic oil pump in the present application; the structure of the electronic oil pump of the fourth embodiment is described in detail below.

(49) Referring to FIG. **20** and FIG. **21**, in this embodiment, the electronic oil pump **100c** includes a conductive member **9**, the conductive member **9** electrically connects the reference ground plane of the electric control board assembly **6** with the first housing **7**, and the first housing **7** can conduct electricity; specifically, the conductive member **9** includes a first part **911** and a second part **912**, the first part **911** is electrically connected with the reference ground plane of the electric control board assembly **6**, the second part **912** abuts against the inner wall of the first housing **7**, the reference ground plane of the electric control board assembly **6** is electrically conducted with the first housing **7** through the conductive member **9**; the electronic oil pump **100c** further includes a shielding member **10**, the shielding member **10** is located between the main body **51** of the partition member **5** and the stator assembly **4**, the shielding member **10** is fixedly connected with the partition member **5**, the shielding member **10** is in contact with the conductive member **9**, as part of the conductive member **9** is electrically connected with the reference ground plane of the electric control board assembly **6**, another part of the conductive member **9** contacts the first housing **7**, so that the shielding member **10** may also be indirectly connected with the reference ground plane of the electric control board assembly **6**; in the above manner, the first housing **7** and the shielding member **10** are both indirectly electrically connected with the reference ground plane of the electric control board assembly **6**; the beneficial effects are: firstly, a low impedance path is provided for the interference voltage in the stator assembly **4** and the electric control board assembly **6**, which is beneficial to reducing the electromagnetic radiation of the stator assembly **4** and the electric control board assembly **6**, and further reducing the interference of electromagnetic radiation generated by the stator assembly **4** and the electric control board assembly **6** on electronic components/or other

external devices; secondly, when the surface of the first housing **7** and the shielding member **10** has static electricity, the static electricity can be conducted to the reference ground plane of the electric control board assembly **6**, when the reference ground plane of the electric control board assembly **6** is grounded to the outside, the static electricity conducted to the reference ground plane is further conducted to the outside grounding point, which is conducive to reducing the static electricity accumulated on the surface of the first housing **7** and the shielding member **10**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components; thirdly, when the surface of the electric control board assembly **6** has static electricity, the above static electricity can be conducted to the first housing **7** and the shielding member **10** through the conductive member **9**, which is conducive to reducing the static electricity accumulated on the surface of the electric control board assembly **6**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components; besides, in this embodiment, the stator core **41**, the second housing **8** and the pump cover **1** are all in contact with the first housing **7**, on one hand, the stator core **41** and the second housing **8** are electrically connected with the reference ground plane of the electric control board assembly, the static electricity on the surface of the stator core **41**, the second housing **8** and the pump cover **1** can be conducted to the reference ground plane of the electric control board assembly **6**, the above static electricity is further conducted to the outside grounding point, which is conducive to reducing the influence of static electricity on the electric control board assembly **6** and/or other external devices; on the other hand, the grounding area of the electronic oil pump can be increased, which can provide a low impedance path for the interference voltage in the electric control board assembly **6** and the stator assembly **4**, which is beneficial to reduce an electromagnetic radiation of the electric control board assembly **6** and the stator assembly **4**, and further to reduce the interference of the above electromagnetic radiation to electronic components and/or other external devices; in this embodiment, the pump cover **1**, the first housing **7** and the second housing **8** are made of metal, of course, it is also possible that the material of the first housing **7** is metal, and the material of the pump cover **1** and the second housing **8** is nonmetal. Referring to FIG. **1**, in this embodiment, static electricity is conducted by one end of the conductive member **9** contacting the first housing **7**, but of course, one end of the conductive member **9** may also contact the stator core **41** or the second housing **8**.

(50) Compared with the first embodiment of the electronic oil pump, in this embodiment, the electronic oil pump **100c** further includes a shielding member **10**, which is in contact with the conductive member **9**; the conductive member in this embodiment can refer to the conductive member in the first embodiment of the electronic oil pump, the shielding member in this embodiment can refer to the shielding member in the second embodiment of the electronic oil pump, which is not repeated here.

(51) Referring to FIG. **22**, FIG. **22** is a structural diagram of a fifth embodiment of an electronic oil pump in the present application; the structure of the electronic oil pump of the fifth embodiment is described in detail below.

(52) Referring to FIG. **22**, in this embodiment, the electronic oil pump **100d** includes a conductive member **9**, the conductive member **9** electrically connects the reference ground plane of the electric control board assembly **6** with the first housing **7**, and the first housing **7** can conduct electricity; specifically, the conductive member **9** includes a first part **911** and a second part **912**, the first part **911** is electrically connected with the reference ground plane of the electric control board assembly **6**, the second part **912** abuts against the inner wall of the first housing **7**, the reference ground plane of the electric control board assembly **64** is electrically conducted with the first housing **7** through the conductive member **9**; the electronic oil pump **100c** further includes a shielding member **10**, the shielding member **10** is located between the main body **51** of the partition member **5** and the electric control board assembly **6**, the shielding member **10** is fixedly connected with the partition member **5**, the shielding member **10** is in contact with the conductive member **9**, as part of the

conductive member **9** is electrically connected with the reference ground plane of the electric control board assembly **6**, another part of the conductive member **9** contacts the first housing **7**, so that the shielding member **10** may also be indirectly connected with the reference ground plane of the electric control board assembly **6**; in the above way, the beneficial effects are: firstly, a low impedance path is provided for the interference voltage in the stator assembly **4** and the electric control board assembly **6**, which is beneficial to reducing the electromagnetic radiation of the stator assembly **4** and the electric control board assembly **6**, and further reducing the interference of electromagnetic radiation generated by the stator assembly **4** and the electric control board assembly **6** on electronic components/or other external devices; secondly, when the surface of the first housing **7** and the shielding member **10** has static electricity, the static electricity can be conducted to the reference ground plane of the electric control board assembly **6**, when the reference ground plane of the electric control board assembly **6** is grounded to the outside, the static electricity conducted to the reference ground plane is further conducted to the outside grounding point, which is conducive to reducing the static electricity accumulated on the surface of the first housing **7** and the shielding member **10**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components; thirdly, when the surface of the electric control board assembly **6** has static electricity, the above static electricity can be conducted to the first housing **7** and the shielding member **10** through the conductive member **9**, which is conducive to reducing the static electricity accumulated on the surface of the electric control board assembly **6**, and further helps to reduce the influence of the above static electricity on the performance of external systems and/or electronic components.

(53) Compared with the first embodiment of the electronic oil pump, in this embodiment, the electronic oil pump **100d** further includes a shielding member **10**, which is in contact with the conductive member **9**; the conductive member in this embodiment can refer to the conductive member in the first embodiment of the electronic oil pump, the shielding member in this embodiment can refer to the shielding member in the third embodiment of the electronic oil pump, which is not repeated here.

(54) The above embodiments are only used to illustrate the present application rather than limit the technical solutions described in the present application. Although the present application is described in detail in this specification with reference to the above embodiments, those of ordinary skill in the art should understand that those skilled in the art may still modify or equivalently replace the present application, and all technical solutions and improvements thereof that do not depart from the spirit and scope of the present application shall be covered within the scope of the claims of the present application.

## Claims

1. An electronic oil pump, comprising a stator assembly and an electric control board assembly, wherein the stator assembly is electrically connected with the electric control board assembly; the electric control board assembly comprises an electronic component; the electronic oil pump further comprises a pump housing and a conductive member, the pump housing can conduct electricity, and the conductive member electrically connects a reference ground plane of the electric control board assembly with the pump housing, wherein the electronic oil pump further comprises a partition member, at least part of the partition member is located between the stator assembly and the electric control board assembly; wherein the conductive member is used as an insert, and the partition member is formed by injection molding, along a height direction of the partition member, part of the conductive member passes through the partition member; wherein the joint between the conductive member passing through the partition member and the partition member is sealed, wherein the partition member comprises a first convex part, the first convex part protrudes from an upper surface of a main body of the partition member, and the conductive member passes through

the first convex part: wherein the first convex part comprises a groove, and the groove is recessed from the upper surface of the first convex part, the groove does not reach the lower surface of the main body of the partition member, the conductive member passing through the main body of the partition member passes through the groove, sealant is filled between the outer periphery of part of the conductive member and the inner wall of the groove, and this part of the conductive member is located in the groove.

2. The electronic oil pump according to claim 1, wherein the pump housing comprises a first housing, the first housing is at least partially arranged on the periphery of the stator assembly; the conductive member comprises a first part and a second part, the first part is electrically connected with the reference ground plane of the electric control board assembly, the second part abuts against the inner wall of the first housing; the reference ground plane of the electric control board assembly is electrically conducted with the first housing through the conductive member.

3. The electronic oil pump according to claim 2, wherein the electric control board assembly comprises a substrate, the substrate comprises a first surface and a second surface, the first surface is closer to the stator assembly than the second surface; wherein the electric control board assembly further comprises an abutting part, the abutting part is located on the first surface, and the abutting part is electrically connected with the reference ground plane of the electric control board assembly, the first part is in contact with the abutting part and electrically connected with the abutting part.

4. The electronic oil pump according to claim 2, wherein the electric control board assembly comprises a substrate, the substrate comprises a first surface and a second surface, the first surface is closer to the stator assembly than the second surface; the substrate comprises a through hole, the through hole is defined in a thickness direction of the substrate, the first part inserts into the through hole and is tightly matched with the through hole.

5. The electronic oil pump according to claim 2, wherein the conductive member is at least partially elastic, and the elastic conductive member comprises an inclined part, the front end of the inclined part is connected with the second part, or, the second part is located at the front end of the inclined part, the inclined part is integrally provided with the second part; the inclined part is disposed at an angle with the inner wall of the first housing.

6. The electronic oil pump according to claim 5, wherein the conductive member comprises a first conductive part and a second conductive part, the first conductive part and the second conductive part are arranged separately, the first conductive part and the second conductive part are in contact and connected; the second conductive part has elasticity, the first part is formed on the first conductive part, and the second part is formed on the second conductive part.

7. The electronic oil pump according to claim 6, wherein the second conductive part comprises a body part and a limiting part, one end of the limiting part is connected with one side surface of the body part, the other end of the limiting part is connected with the other side surface of the body part, the inclined part is connected with the upper end of the body part, the second part is located at one side of the body part, and part of the limiting part is located at the other side of the body part; wherein the limiting part is provided with an accommodating cavity, the first conductive part passes through the accommodating cavity, a first end of the first conductive part is located at one side of the limiting part, and a second end of the second conductive part is located at the other side of the limiting part; wherein the first conductive part is in contact with the body part.

8. The electronic oil pump according to claim 7, wherein the limiting part further comprises a first sub-part and a second sub-part, a first end of the first sub-part is connected with the first side surface of the body part, a second end of the first sub-part is a free end, and a first end of the second sub-part is connected with the second side surface of the body part, a second end of the second sub-part is a free end, and the second end of the first sub-part has a set distance from the second end of the second sub-part.

9. The electronic oil pump according to claim 6, wherein the stator assembly comprises a stator

core, a winding and an insulating frame, the insulating frame covers at least part of the surface of the stator core, and the winding is wound around the insulating frame, the first end of the first conductive part is electrically connected with the reference ground plane of the electric control board assembly, the second end of the first conductive part is connected with the insulating frame, the first conductive part is non-electrically connected with the winding.

10. The electronic oil pump according to claim 1, wherein the electronic oil pump comprises a shielding member, the stator assembly, at least part of the shielding member and the electric control board assembly are distributed along the height direction of the electronic oil pump, at least part of the shielding member is located between the stator assembly and the electric control board assembly; the shielding member is made of conductive metal material, and at least part of the shielding member covers at least part of the winding of the stator assembly.

11. The electronic oil pump according to claim 10, wherein the electronic oil pump further comprises a partition member, at least part of the partition member is located between the stator assembly and the electric control board assembly; the shielding member is connected with the partition member.

12. The electronic oil pump according to claim 11, wherein the partition member further comprises at least two protruding posts, along the axial direction of the partition member, the at least two protruding posts are convexly arranged from the end face of a main body of the partition member; the shielding member comprises at least two through holes, and the at least two protruding posts pass through the through holes, the shielding member and the partition member can be fixedly connected by hot riveting the part of the at least two protruding posts protruding out of the through hole.

13. The electronic oil pump according to claim 11, wherein the conductive member comprises a first part, the first part is electrically connected with the reference ground plane of the electric control board assembly, the shielding member is in contact with the conductive member, the conductive member is electrically connected with the reference ground plane of the electric control board assembly and the shielding member.

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